

AIMAR TRASMONTE DOMENECH

Industrial Engineer

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aimar-trasmonte



Industrial Engineer with experience in industrial machinery design (welded sheet metal, FEA) and digital fabrication. I am especially passionate about design, manufacturing and optimisation. Outside engineering, music and associations have taught me teamwork and leadership through managing projects with many stakeholders.

EXPERIENCE

Mechanical Design Intern

AUSA - CDEI

September 2025 - July 2026

Support to the R&D team of the **Forklifts** division:

- Design and updating of 50+ welded sheet metal and cast assemblies using CREO. Cloud PLM management with Windchill.
- Preparation and updating of part and assembly drawings for manufacturing and homologation.
- FEA (ANSYS and CREO Simulate) of failures in testing and in service; design of reinforcements to eliminate the failures.

Digital Fabrication Lab Technician

EngiLab ETSEIB

January 2025 - July 2025

In charge of the open fabrication lab, with 10+ filament (FDM/FFF) and resin (SLA/DLP) 3D printers, an industrial laser cutter and a vinyl cutter, among others:

- Managed 300+ orders from students, departments and others.
- Mechanical design and DFM advice for clients.

Risk and Crisis Management Consulting Intern

Institut Cerdà

February 2024 - June 2024

Preparation of corporate-level risk and crisis prevention reports for large companies. Advice on regulations. Occupational safety benchmarking and accident analysis.

Organisation of the Festa d'Industrials (unpaid)

ETSEIB Student Assembly

Annually since 2021

Event the street for 2,000 people: hiring suppliers, handling permit paperwork and coordinating staff.

EDUCATION

Master's Degree in Industrial Engineering – Mechanical Specialisation

UPC - ETSEIB

2024 – January 2027 (expected)

- Relevant courses: Mechanical Design, Machine Testing and Design, Vibrations, Bolted Joints, Part Manufacturing.
- Master's Thesis: Design of a full-stack web platform for the dissemination of musical content. In progress.

Bachelor's Degree in Industrial Technologies Engineering

UPC - ETSEIB

2020 – 2024

- Bachelor's Thesis: Design and construction of a competition RC sailboat. Graded with Honours.
- Winner of the degree's *Viu la UPC* award. (recognition of student involvement)
- Erasmus exchange at Politechnika Wrocławska (Wrocław, Poland).

LANGUAGES

Catalan – Native

English – C1

Spanish – Native

Italian – A2

TECHNICAL PROJECTS

RC Sailboats - RC Sailing Barcelona race



Runner-up – 2024 & 2025 Editions

Design and construction of 2 RC sailboats with 3D printing and composites (fiber glass & epoxy). Started as a Bachelor's Thesis (Honours).



Marimba via modal FEA optimisation



Design and construction of a marimba (xylophone), tuning the bars through iterative geometry optimisation with modal FEA.

Electronics and programming



Arduino projects (LED control via BT with an Android app) and full-stack web apps with React, Node and Next.

OTHER PROJECTS

Founding member of Cadència Industrial



2021 – Present

The ETSEIB musicians' association.

ETSEIB student representative



2022 - 2024

Elected on the school board and member of the student assembly.

SKILLS

CAD Software

CREO Elements and Parametric

SolidWorks

Windchill (PLM)

AutoCAD

Simulation and calculation

ANSYS (FEA)

MATLAB

Python

Design and manufacturing

Welded sheet metal

ISO drawings

GD&T

DFM

3D printing

Laser cutting

Other

Illustrator

LaTeX

Office / Google Workspace

OTHER QUALIFICATIONS



Car driving licence (B) - Own vehicle



Recreational Boat Skipper (PER)



Professional Music Degree - Violin